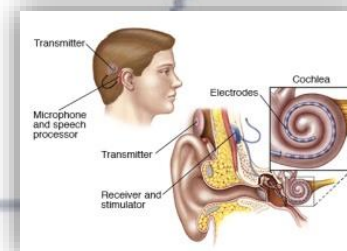
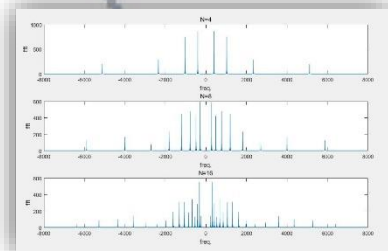
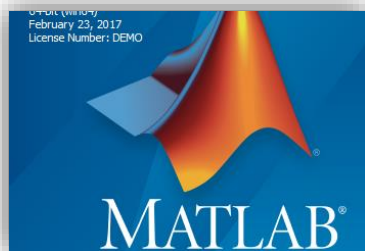


信号与系统实验

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Signals and Systems (Lab)

Lab 3: Fourier Series Representation of Periodic Signals

Part 2

Dr. **Wu Guang**

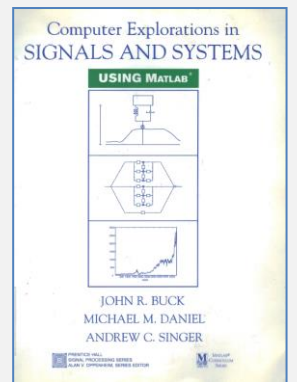
wug@sustech.edu.cn

Electrical & Electronic Engineering
Southern University of Science and Technology

Labs in this course

5 Lab assignments+2 Projects

- Lab 1: MATLAB Programming (3 Weeks)
- Lab 2: Linear Time-Invariant Systems (2 Weeks)
- **Lab 3: Fourier Series Representation of Periodic Signals (2 Weeks)**
- Lab 4: The Continuous-Time Fourier Transform (2 Weeks)
- Lab 5: Coding Test & System, Transform, Convolution and Filter (1 Week)
- Project 1: Speech synthesis and perception with envelope cue (3 Weeks)
- Project 2: Motion detection via communication signals (3 Weeks)



Review—Complex Exponentials

$$\begin{aligned} x(t) = e^{st} &\longrightarrow \boxed{h(t)} \longrightarrow y(t) = \int_{-\infty}^{+\infty} h(\tau) e^{s(t-\tau)} d\tau \\ &= \left[\int_{-\infty}^{+\infty} h(\tau) e^{-s\tau} d\tau \right] e^{st} \\ &= \underbrace{H(s)}_{\text{eigenvalue}} \underbrace{e^{st}}_{\text{eigenfunction}} \end{aligned}$$

$s = j\omega$ - purely imaginary,
i.e. signals of the form $e^{j\omega t}$



The system H is LTI

Review—Exercise 1: Frequency response

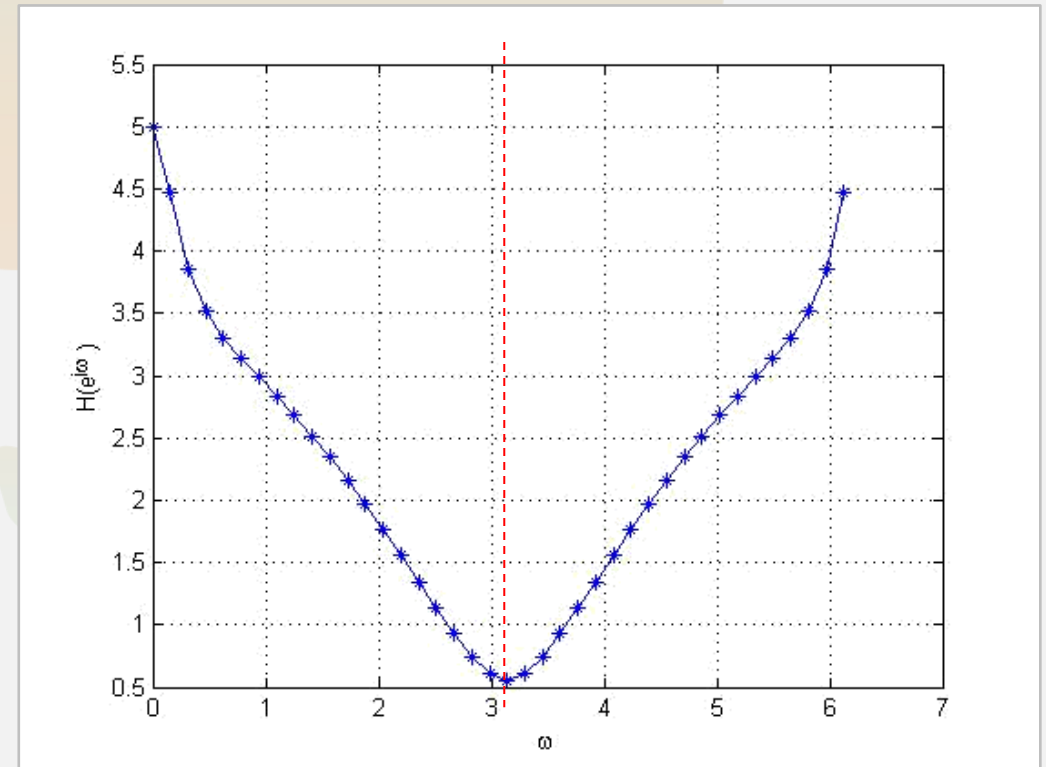
➤ Consider LTI System: $y[n] - 0.8y[n - 1] = 2x[n] - x[n - 2]$

Define the vector of coefficients:

```
A=[1 -0.8];  
B=[2 0 -1];
```

Plot the frequency response:

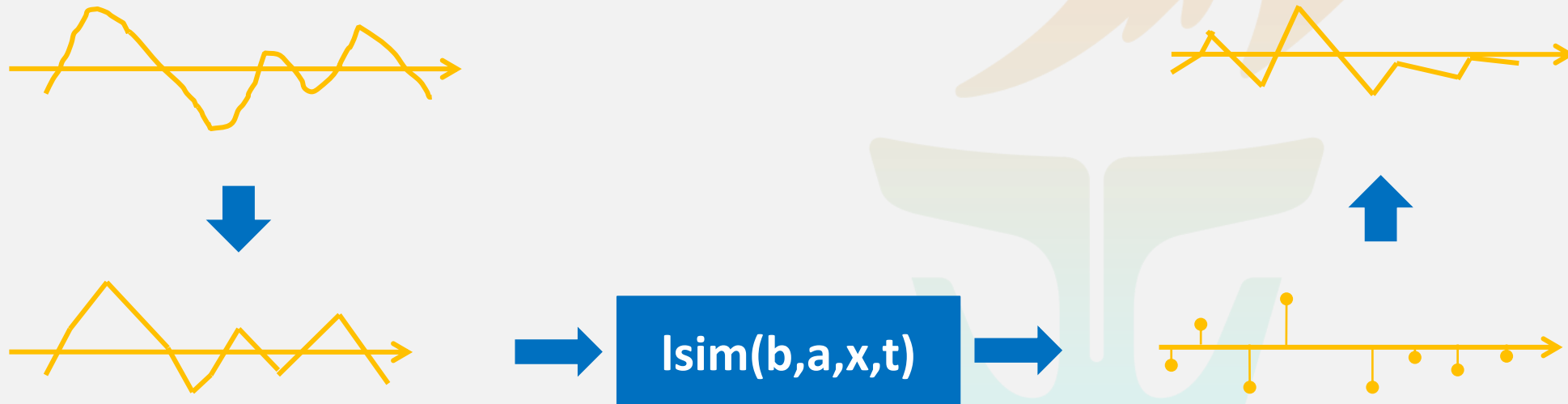
```
[H Omega] = freqz(B, A, 40, 'whole');  
plot(Omega, abs(H), '*-');  
xlabel('\omega');  
ylabel('H(e^{j\omega})');  
grid;
```



Review—CT LTI System: Isim

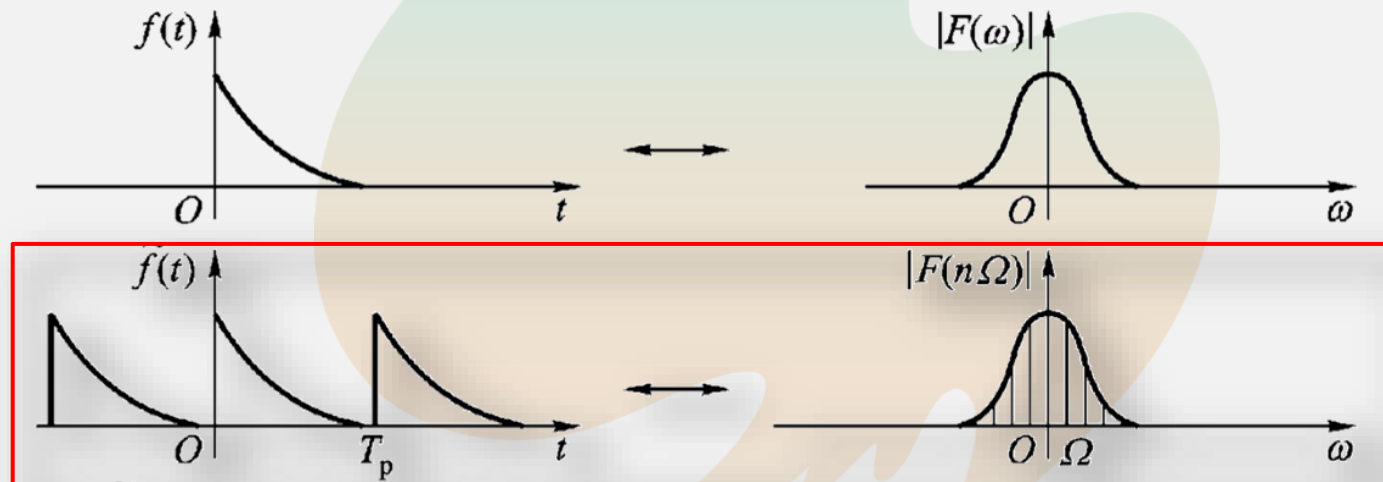
- **Isim()**: generate sampled output according to sampled input signal and CT system function
- **Syntax**: `Isim(b,a,x,t)`

Sampled input signal
Vector of sampling time: t
Vector of sampled value: x

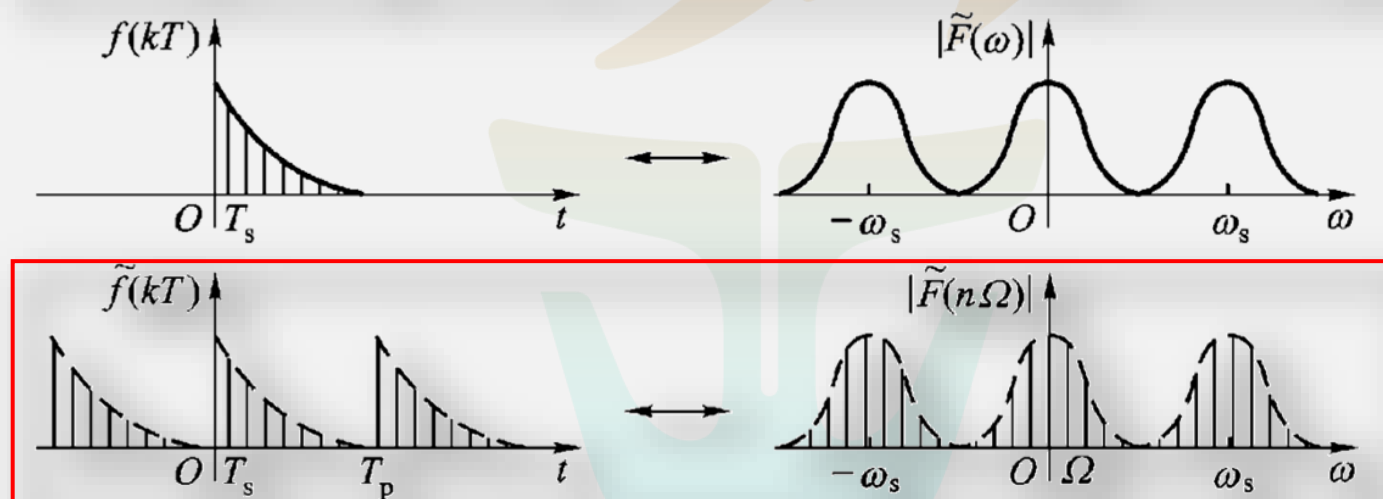


1. Q: How to calculate DTFS of a periodic signal

FS
Fourier Series



DTFS
Discrete Time Fourier Series



2.1 Fourier Series of Periodic Signal

- Periodic signal with period T or N
- Synthesis equation:

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\left(\frac{2\pi}{T}\right)t} \quad \text{v.s.} \quad x[n] = \sum_{k=0}^{N-1} a_k e^{jk\left(\frac{2\pi}{N}\right)n}$$

- Analysis equation:

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\left(\frac{2\pi}{T}\right)t} dt \quad \text{v.s.} \quad a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk\left(\frac{2\pi}{N}\right)n}$$

2.2 MATLAB function: fft

- `fft()`: compute DTFS coefficients from signals

```
>> help fft
```

$$X(k) = \sum_{n=1}^N x(n) \exp(-j \cdot 2 \cdot \pi \cdot (k-1) \cdot (n-1) / N), \quad 1 \leq k \leq N.$$

- Compare with our definition:

$$a_k = \sum_{n=1}^N x[n] e^{-j(k-1)\left(\frac{2\pi}{N}\right)(n-1)}$$

v.s.

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk\left(\frac{2\pi}{N}\right)n}$$

- Calculate the DTFS of vector `x`:

$$a = (1/N) * \text{fft}(x)$$

2.3 MATLAB Function: ifft

- ifft(): reconstruct signals from DTFS coefficients

```
>> help ifft
```

$$x(n) = \frac{1}{N} \sum_{k=1}^N X(k) \exp(j \cdot 2 \cdot \pi \cdot (k-1) \cdot (n-1) / N), \quad 1 \leq n \leq N.$$

- Compare with our definition:

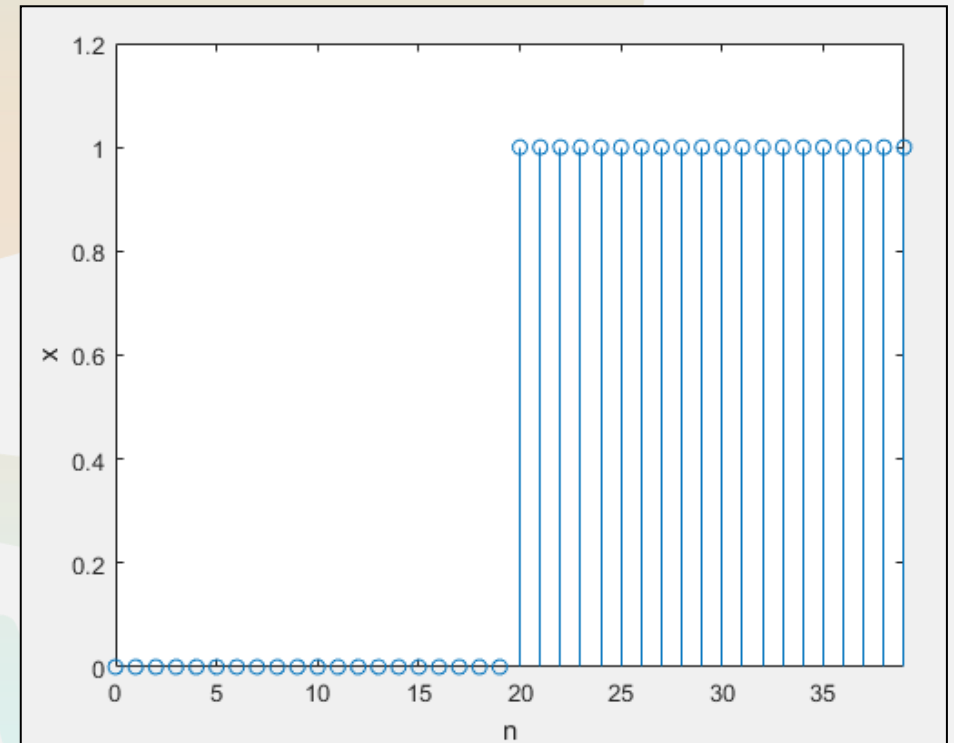
$$x[n] = \frac{1}{N} \sum_{k=1}^N a_k e^{j(k-1)\left(\frac{2\pi}{N}\right)(n-1)} \quad \text{v.s.} \quad x[n] = \sum_{k=0}^{N-1} a_k e^{jk\left(\frac{2\pi}{N}\right)n}$$

- Calculate the signal x: $x = N * \text{ifft}(a)$

2.4 Exercise—fft

- Periodic DT rectangular wave with **period = 40**

```
x=zeros(1,20)  
ones(1,20);  
stem(0:39, x);  
xlim([0 39]);  
ylim([0 1.2]);
```



2.4 Exercise—fft

```
A = fft(x) / length(x);  
figure(1)  
subplot(2,1,1),stem(abs(A));  
subplot(2,1,2),stem(abs(fftshift(A)));
```

```
>> fftshift([1 2 3 4 5 6])
```

```
ans =
```

```
4      5      6      1      2      3
```

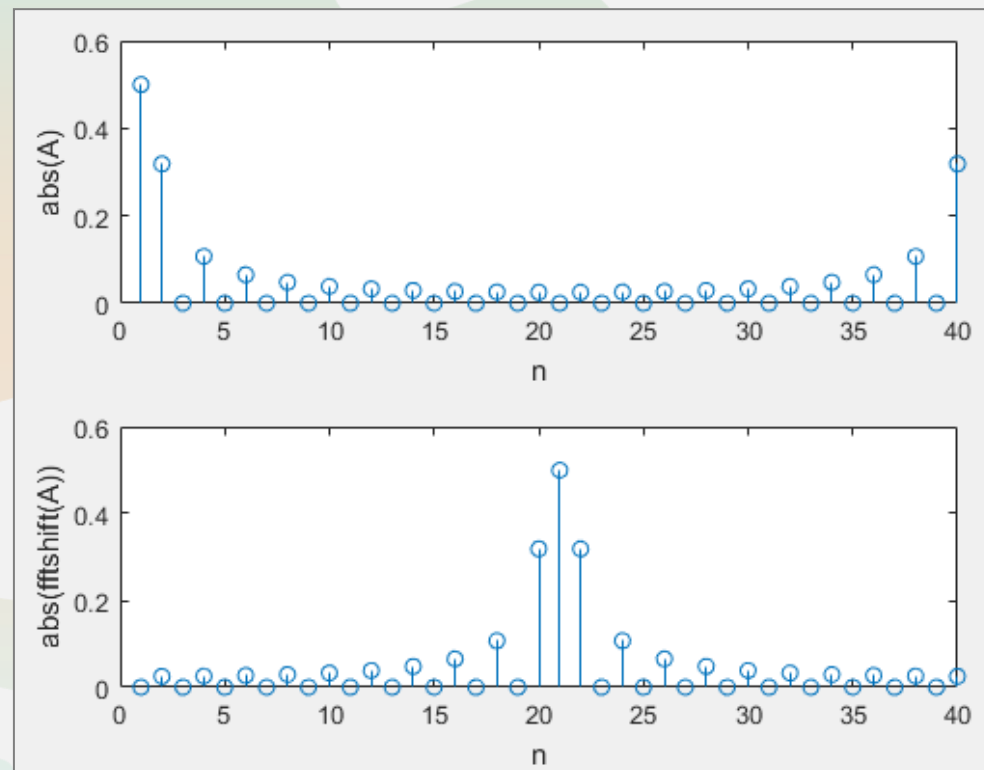
```
>> help fftshift
```

fftshift - 将零频分量移到频谱中心

此 MATLAB 函数 通过将零频分量移动到数组中心，重新排列傅里叶变换 x 。

```
Y = fftshift(X)
```

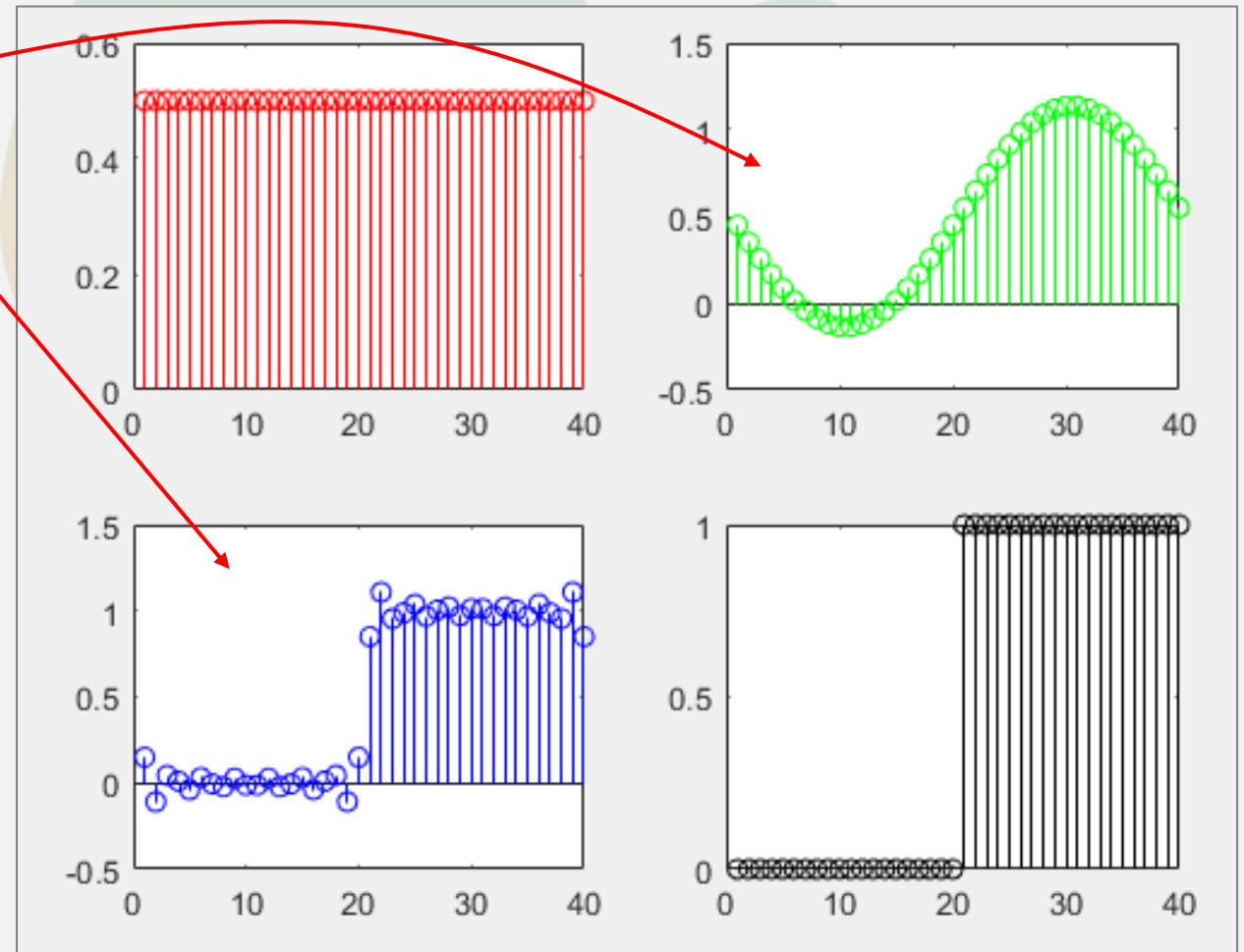
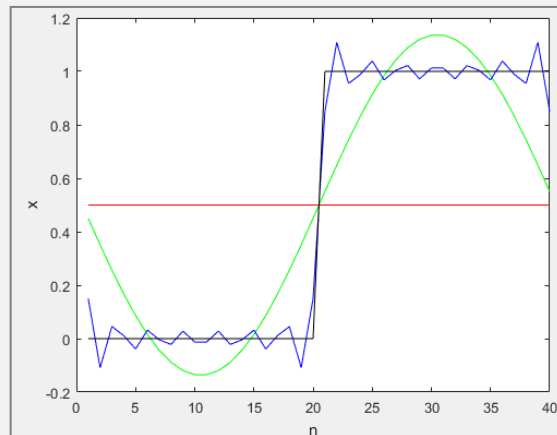
```
Y = fftshift(X,dim)
```



2.5 Exercise—ifft

```
A1 = [A(1) zeros(1,39)];  
A2 = [A(1) A(2) zeros(1,37) A(40)];  
A3 = [A(1) A(2:15) zeros(1,11) A(27:40)];  
subplot(2,2,1), stem(1:40,ifft(A1)*40, 'r');  
subplot(2,2,2), stem(1:40,ifft(A2)*40, 'g');  
subplot(2,2,3), stem(1:40,ifft(A3)*40, 'b');  
subplot(2,2,4), stem(1:40,x, 'k');
```

Reconstruct signals



3.1 Complexity Analysis

- Suppose we know the matrix

$$E(n, k) = e^{-jk\left(\frac{2\pi}{N}\right)n}$$

- How many multiplications & additions are needed to calculate Fourier series $[a_0, a_1, \dots, a_{N-1}]$

$$a_0 = \frac{1}{N} \sum_{n=0}^{N-1} x[n]E(n, 0)$$

...

$$a_{N-1} = \frac{1}{N} \sum_{n=0}^{N-1} x[n]E(n, N-1)$$

Each: $N+1$ $\times$; $N-1$ $+$
Total: $(N+1)N$ $\times$; $N(N-1)$ $+$

3.2 Complexity Analysis

➤ Fast Fourier Transform:

Calculation of Fourier series (transform) can be speeded up
Complexity reduces to $O(N\log N)$

$$N=4$$

$$a_0=(x[0]E(0,0)+x[1]E(1,0)+x[2]E(2,0)+x[3]E(3,0))/N$$

$$a_2=(x[0]E(0,2)+x[1]E(1,2)+x[2]E(2,2)+x[3]E(3,2))/N$$

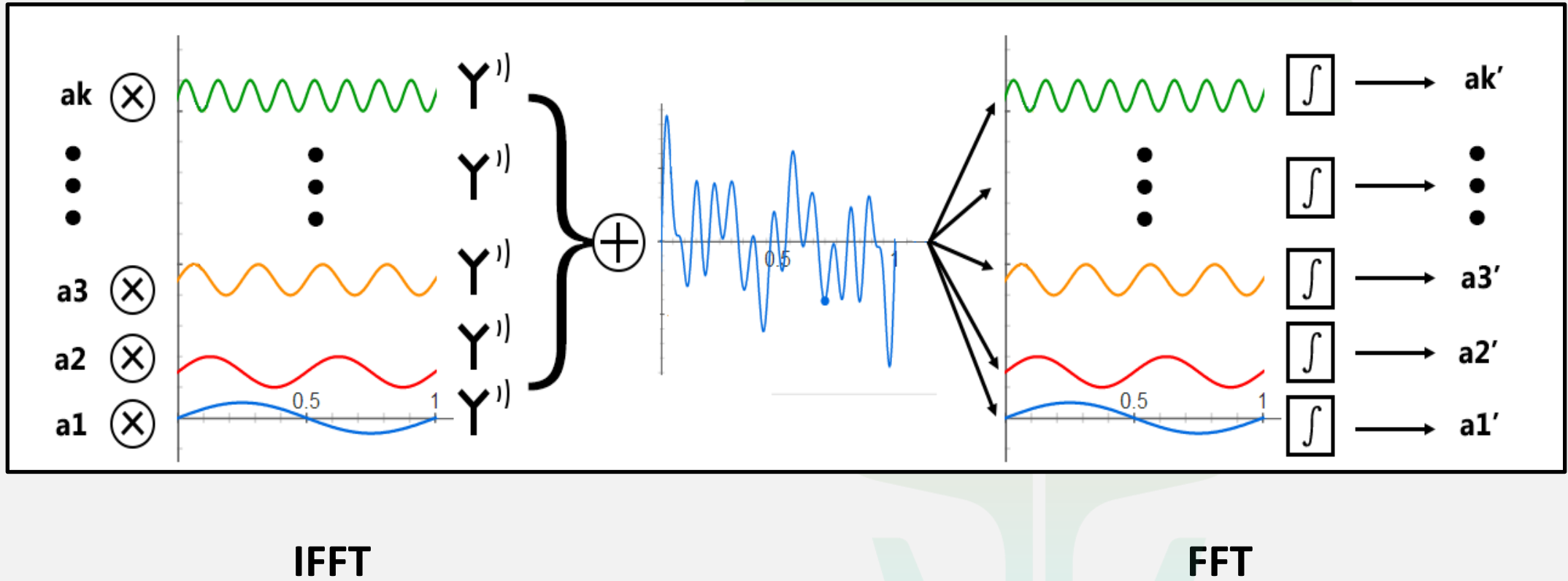
$$a_1=(x[0]E(0,1)+x[1]E(1,1)+x[2]E(2,1)+x[3]E(3,1))/N$$

$$a_3=(x[0]E(0,3)+x[1]E(1,3)+x[2]E(2,3)+x[3]E(3,3))/N$$

3.3 DFT and FFT of a 1024*1024 image

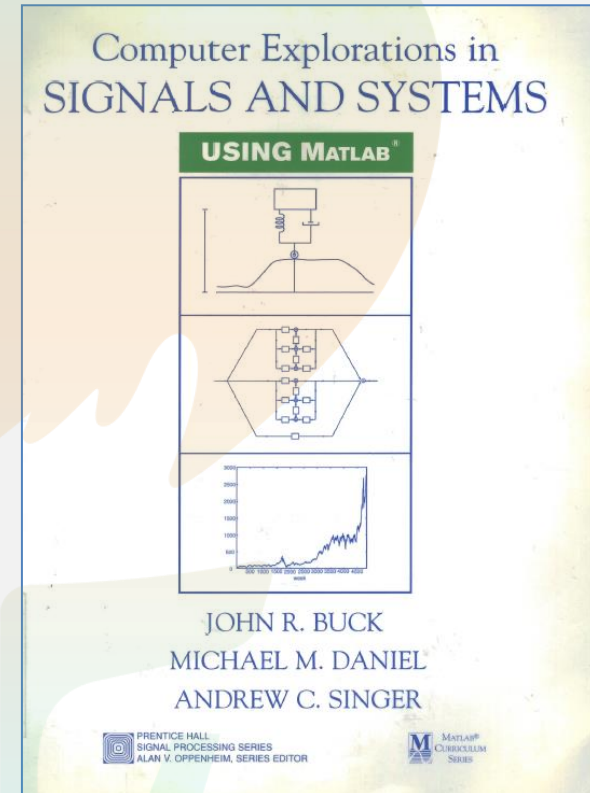
CPU	Clock Frequency	DFT	FFT
1941	60 Hz	152.3 y	271.4 d
1971 (4004)	108KHz	30.8 d	3.6 h
1978 (8086)	10MHz	8.0 h	2.3 min
1982 (80286)	20MHz	4.0 h	1.2min
1985 (80386)	33MHz	2.4h	42.6s
1989 (80486)	100MHz	48.0min	14.1s
1995 (Pentium)	200MHz	24.0min	7.0s
1999 (Pentium III)	450MHz	10.7min	3.1s
2000 (Pentium 4)	1.4GHz	3.4min	1.0s
2001 (Pentium 4)	2GHz	2.4min	0.7s

4. Application: OFDM



4. Lab Assignment

- Read tutorial 3.2 & 3.3 by yourself
- 3.8、3.10
- Submit your report.



Periodic convolution

- **Periodic convolution** in time domain is equivalent to multiplication in frequency domain

$$x[n] \otimes \hat{h}[n] = \sum_{r=0}^{N-1} x[r] \hat{h}[n-r] \Leftrightarrow N a_k h_k$$

Table 3.2 of Textbook

$$y[n] = x[n] * h[n] = x[n] \otimes \hat{h}[n]$$

$\hat{h}[n]$ is a periodic version of $h[n]$

Question ?

